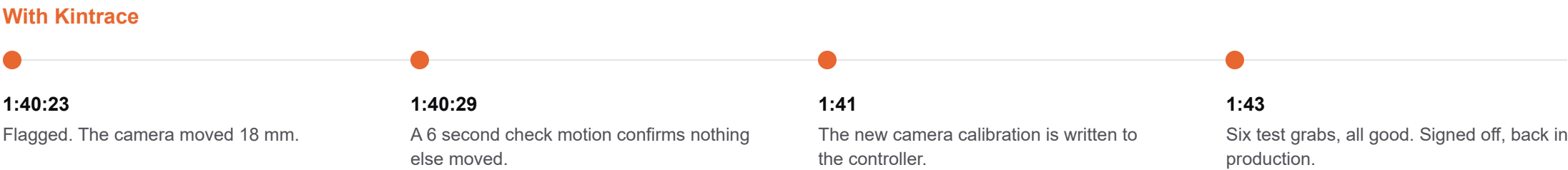
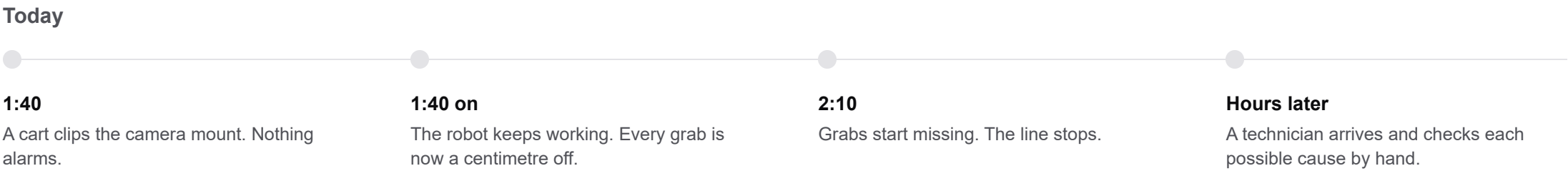

Kintrace

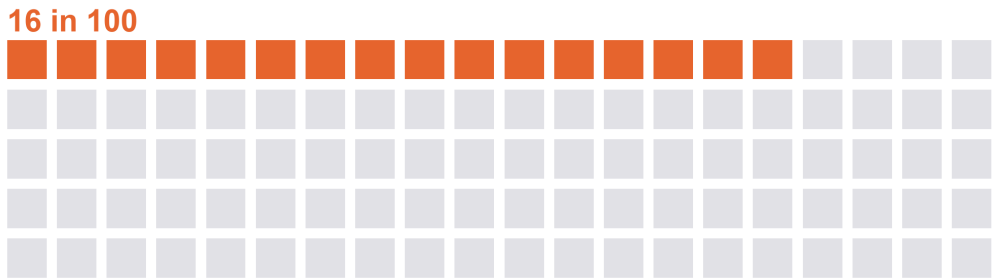
Robots get knocked out of place and nobody notices. Kintrace finds what moved and puts it back.

At 1:40 a cart clips a camera. Nobody finds out until 2:10.



An illustration. The 18 mm and the timings are printed by the tool, running on our simulator. No cell like this exists yet.

You pay for it twice: in ruined training data, and in stopped lines.



Robot training data

An open audit of 4,959 recorded episodes flagged 15.6% of them for exclusion, and another half for review.

\$2.3M
an hour

Up to, for an idle production line at a big car plant. Across the 500 largest manufacturers, unplanned downtime of every kind costs \$1.4 trillion a year, about 11% of revenue.

Everything on the market tells you something is wrong, not what moved.

Monitoring dashboards

Ember, Foxglove, Formant

They watch the software and the logs. They will tell you a cell is slow or a run failed. They do not measure where the camera is, so they cannot name the physical cause.

Calibration hardware

Dynalog, CAPTRON, Renishaw

These do measure, and they are accurate. But you bolt a device into every cell you want covered, and each one mostly does the tool.

A technician

How it is actually done

Someone drives out and rules out each cause by hand, one at a time, with the cell down while they do it. It works. It costs hours.

We have not found anyone doing all of it from software alone: when it changed, what moved, by how much, and the fix written back.

Kintrace reads the robot's own logs, then says what moved and writes the fix.

```
$ KINTRACE INCIDENT                                RECORD.TXT

KINTRACE recommission record, Cell 3
=====
Changed at   : 13:40:23 (lines up with protective stop at 13:40:23: collision detected, arm stopped)
  seen in    : camera vs arm at 13:40:23
  seen in    : camera vs table marker at 13:40:23
  seen in    : first missed pick at 13:40:25

What changed (6 s check motion)
- Camera moved: 18.1 mm, 0.04 deg (parts appear 18 mm off in the pick area)

Fix written to controller
- camera calibration updated

Verification (check motion re-run)
[ok ] Command timing      8 ms vs 8 ms normal
[ok ] Camera vs fixed cell marker table marker where it should be (1.0x noise)
[ok ] Camera vs joint encoders disagreement 1.0x sensor noise
[ok ] Fingertip vs known-good fingertip 0.1 mm from its known-good position
[ok ] Tool offset setting matches known-good
test picks: 6/6 ok

STATUS: GO
```



What we actually did: bump the camera, sized for about 20 mm at the fingertip.

- **Notices**

It finds the moment things changed and lines it up with the protective stop.

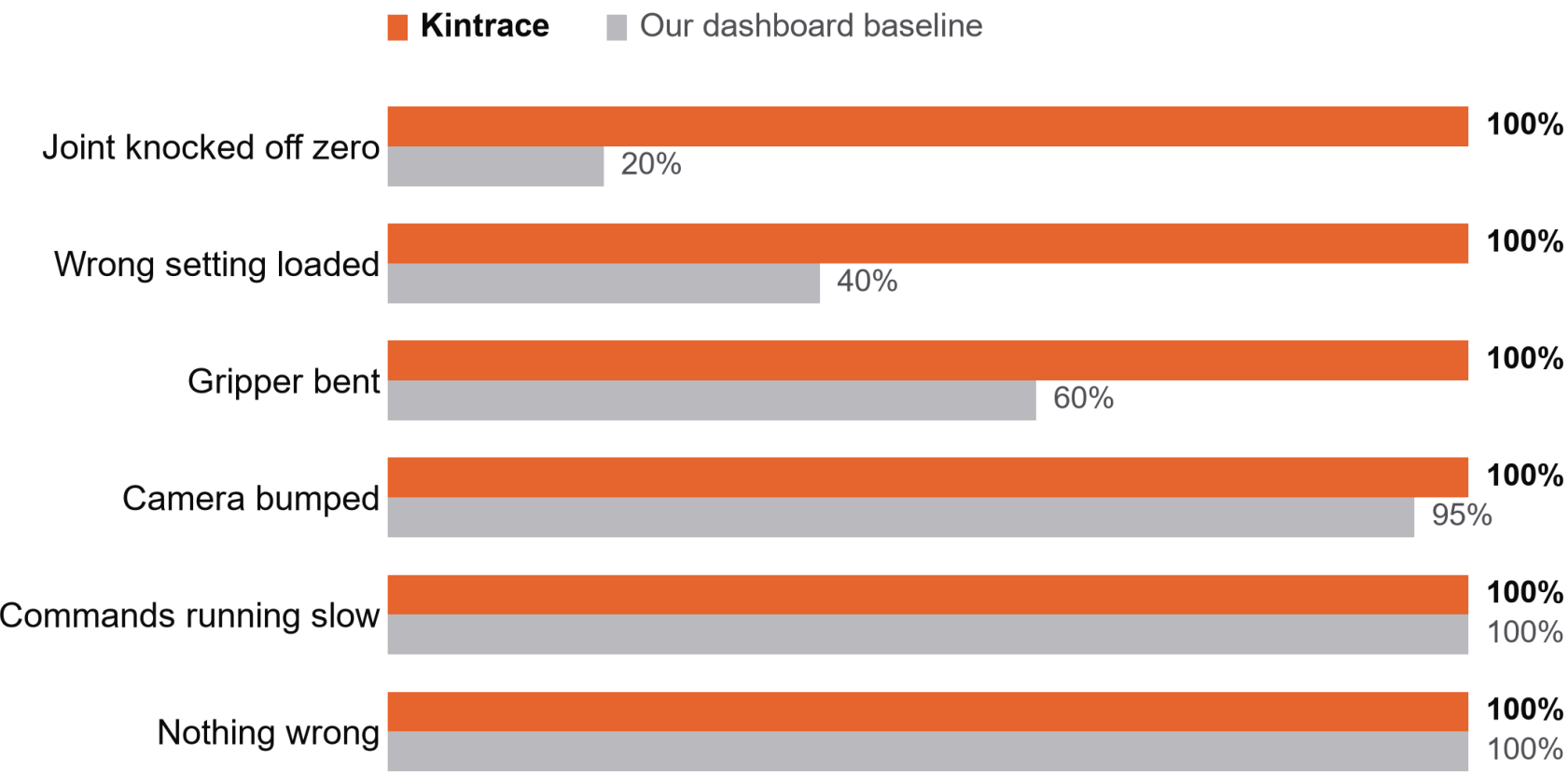
- **Explains**

It names the cause and sizes it. Here: the camera, 18.1 mm.

- **Fixes**

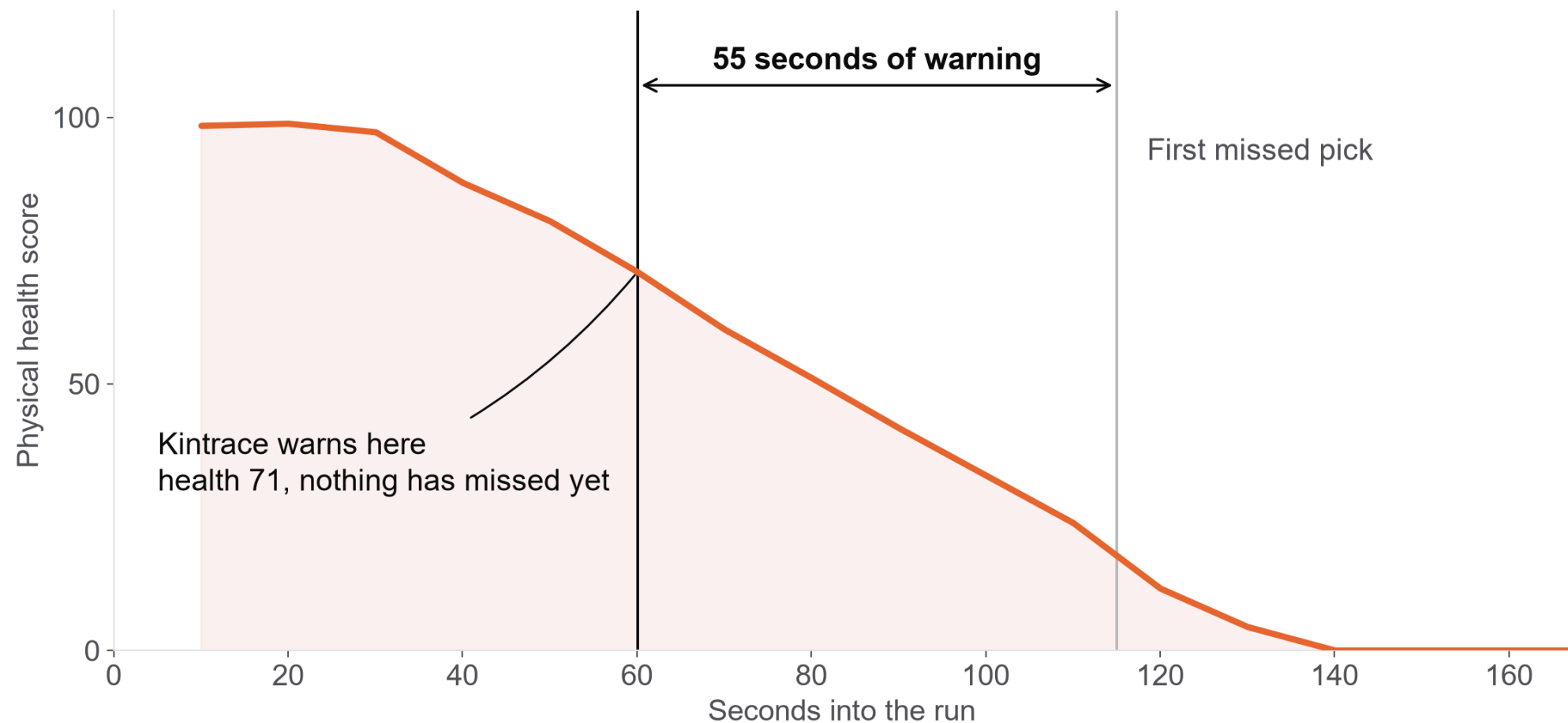
It writes the new calibration, re-checks, runs six test grabs, then signs GO or NO-GO.

In our simulator, Kintrace named the right cause every time. The dashboard baseline we wrote got 69%.



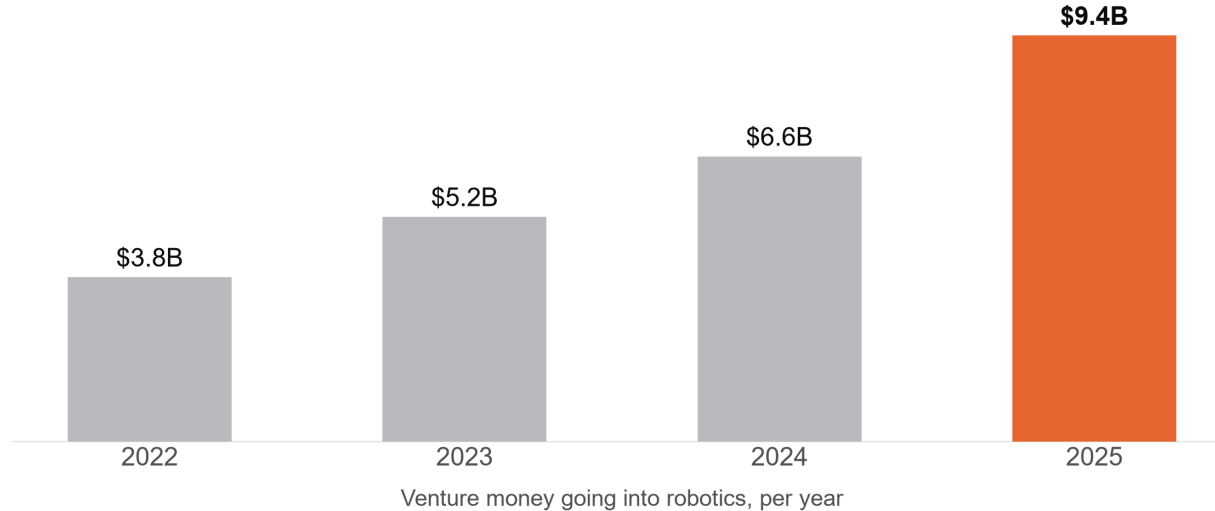
120 runs on our simulated UR5e pick cell, 20 per case, every fault sized to cause the same 20 mm miss. The baseline is ours, built from the signals a typical dashboard tracks. Never run on a real arm.

It also catches slow drift, about 55 seconds before the first grab misses.



One simulated run of a sagging camera mount. Across 20 drifts it warned before the first miss 19 times, named the cause right 20 times, and raised 0 false alarms in 230 healthy windows. Drift is sped up so a run takes minutes. We have not measured lead time at real-world drift rates.

There have never been more robots, and the money behind them doubled in three years.



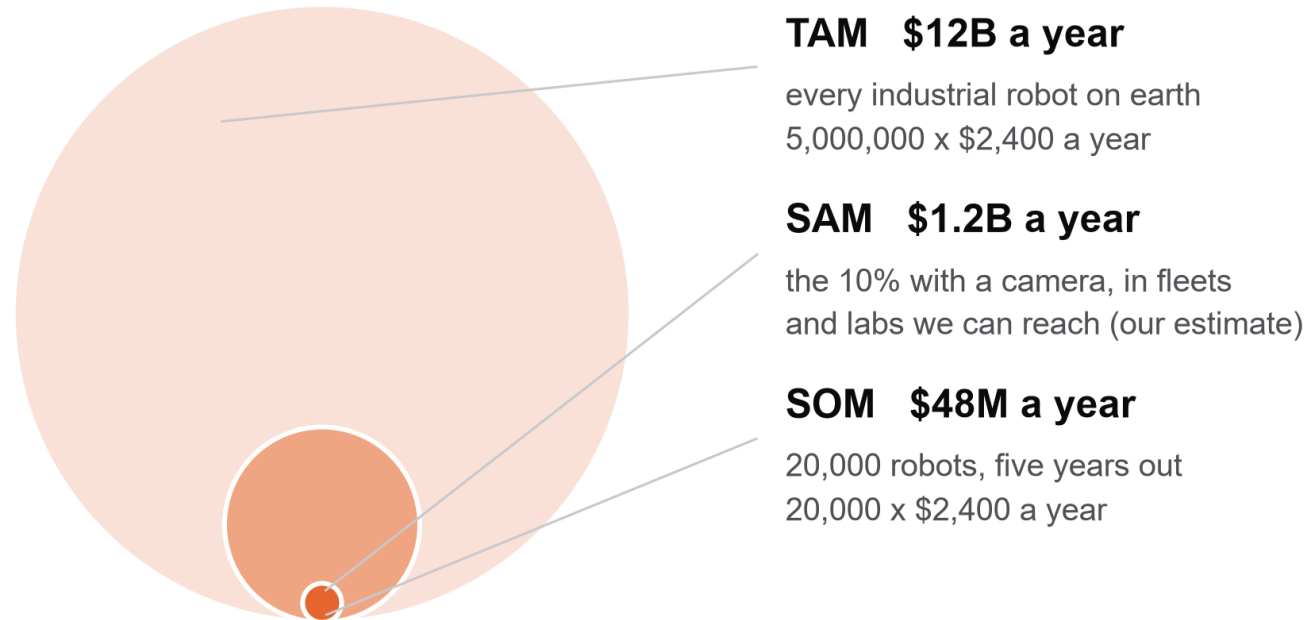
5 million

industrial robots now operating, a record, up 9% in a year. The US installed 38,500 and passed Japan. IFR, September 2026.

47% a year

forecast growth for physical AI, from \$0.89B in 2025 to \$15.2B by 2032. Every robot in it learns from recordings.

At \$200 a robot a month, the part we can realistically sell to is \$1.2B a year.



Top down check: \$22.5B was spent on industrial robot services in 2023, heading to \$41.6B by 2033. Our \$12B bottom up number sits inside a services market that is already twice its size.

\$200 a robot a month, against a technician visit that costs \$1,500 to \$3,000.

How we charge

\$200

per robot, per month, for always-on watching

~\$300

per incident we find and fix. Planned pilot pricing, no pilots yet

\$1,500 to \$3,000

what the technician visit it replaces costs today (our estimate)

Who we sell to, in order

Robot learning teams first, they feel bad data fastest. Then robots-as-a-service fleets, who pay for every callout. Then robot makers.

	Finds and fixes the cause		
Needs hardware	Dynalog, CAPTRON Finds and fixes it, but you mount hardware in every cell.	Kintrace Finds it, sizes it, writes the fix, certifies the cell. Software only.	Software only
	Robot makers Built-in mastering routines. One brand, run by a technician.	Ember, Foxglove, dataset checkers Show you something is off. No geometry, no fix.	
	Only shows a problem		

Raising [\$150K] to put Kintrace on real arms and into three paid pilots.

Where it goes

- A desk arm rig, then time on a lab arm, to reproduce all five faults on real hardware.
- Three paid pilots with robot learning teams.
- A reader so fleets can send their own robot logs, in the formats they already record.

Team

Rithik Satarla

Physical AI research intern at CMU. Harvard Innovation Labs. Wrote all of the above.

[Cofounder]

Next six months

- **Nov 2**
YC W27 application in
- **Month 1-2**
Five faults reproduced on the desk arm, numbers published
- **Month 3-4**
Same five on a lab arm, and the first paid pilot
- **Month 5-6**
Three paid pilots running, real log reader shipped